

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech. (CL) Examination****March - April 2018****CL302.01 Structural Analysis - II****Date: 29.03.2018, Thursday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

Equations:

1. $F\phi = wR \left[\frac{\theta}{2} - \phi \right]$
2. $M\phi = wR^2 \left[\frac{\theta}{2} \sin \phi + \frac{\theta}{2} \cot \frac{\theta}{2} \cos \phi - 1 \right]$
3. $T\phi = wR^2 \left[\frac{\theta}{2} \cos \phi - \frac{\theta}{2} \cot \frac{\theta}{2} \sin \phi - \left(\frac{\theta}{2} - \phi \right) \right]$
4. $\sin \phi = \frac{1}{\frac{\theta}{2}} \left[\sin^2 \frac{\theta}{2} \pm \cos \frac{\theta}{2} \left(\left(\frac{\theta}{2} \right)^2 - \sin^2 \frac{\theta}{2} \right)^{1/2} \right]$

SECTION – I

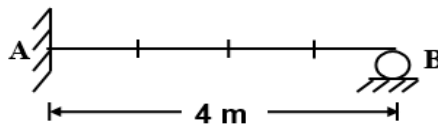
Q - 1 A curved beam circular in plan symmetrically supported on six columns has radius **[07]**
5 m. The beam carries an U.D.L. of 30 kN/m, including self-weight of beam.
Determine S.F., B.M. and T.M. at various locations and draw shear force, bending
moment and twisting moment diagrams. Consider $\frac{\theta}{2} = 30^\circ = 0.5236$ Radian.

Q - 2 (a) Draw qualitative influence line diagrams for 2 equal span continuous beam for **[02]**
reactions at external supports.

(b) Write fixed end moments for a beam having span L and loaded with a concentrated **[02]**
force P acting at midpoint of beam.

(c) Attempt any one question. **[10]**

(i) Draw Influence Line diagram for R_A , R_B and M_A for propped cantilever beam of
length 4 m. Calculate the ordinates at 1 m intervals. (Refer **Figure 1**)

**Figure: 1**

- (ii) Draw Influence Line diagram for R_A and R_C for continuous beam shown in **Figure 2**. Assume EI is constant throughout. Calculate the ordinates at 2 m intervals.

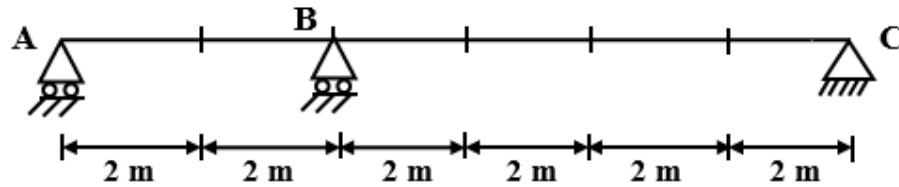


Figure: 2

- Q - 3 (a) Calculate the fixed end moments for the fixed beam shown in **Figure 3**. Also find the support reactions. [07]

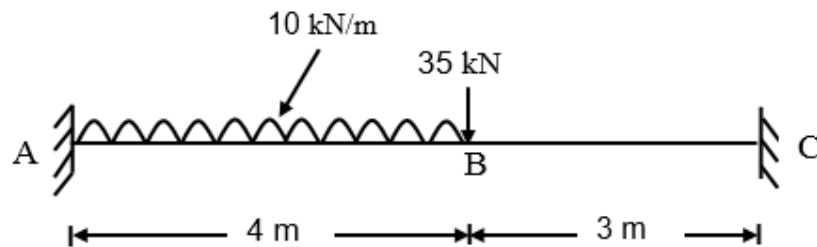


Figure: 3

- (b) A cable is used to support five equal and equidistant loads over a span of 30 m. Find the length of the cable required if the safe tensile stress is 140 N/mm^2 . The central dip of the cable is 2.5 m and loads are 5 kN each. Also find tensions in each part. [07]

OR

- Q - 3 (a) Analyse the continuous beam ABC shown in **Figure 4**. Calculate the support reactions at A, B and C. Also draw the shear force and bending moment diagrams. [07]

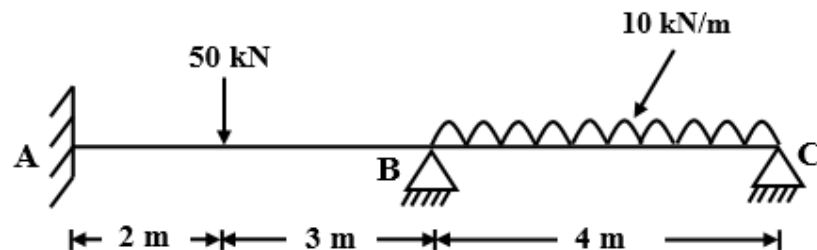


Figure: 4

- (b) A light cable 18 m long is supported at two ends at the same level. The supports are 16 m apart. The cable supports three loads 8, 10 and 12 kN dividing the 16 m distance in four equal parts. Find shape of the cable and tension in various parts. [07]

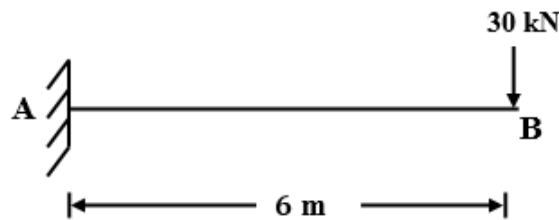
SECTION – II**Q - 4 Do as directed.**

(i) State Castigliano's first theorem. [01]

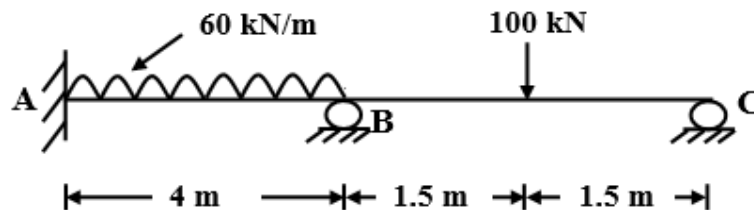
(ii) Define stiffness and flexibility. [02]

(iii) A solid rod is of length 2 m and cross-section 100 mm x 100 mm. If the allowable axial stress in the rod is 100 N/mm^2 , find the magnitude of maximum tensile force that can be applied on the rod suddenly. Take $E = 2 \times 10^5 \text{ N/mm}^2$. Also calculate the strain energy absorbed by the rod at that force. [04]

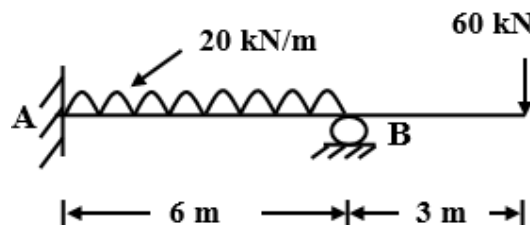
Q - 5 (a) Calculate the deflection at point B for the beam shown in **Figure 5** using Castigliano's first theorem OR Unit load method. Consider $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 3 \times 10^8 \text{ mm}^4$. [04]

**Figure: 5**

(b) Find all reactions for the beam shown in **Figure 6** using Stiffness method OR Flexibility method. Also draw shear force diagram. [10]

**Figure: 6**

Q - 6 (a) Find R_A and R_B for the beam shown in **Figure 7** by Unit Load method OR Castigliano's 2nd theorem. Consider EI as constant for all members. [07]

**Figure: 7**

- (b) Analyse the propped cantilever beam shown in **Figure 8** using consistent deformation method. Draw shear force and bending moment diagrams. [07]

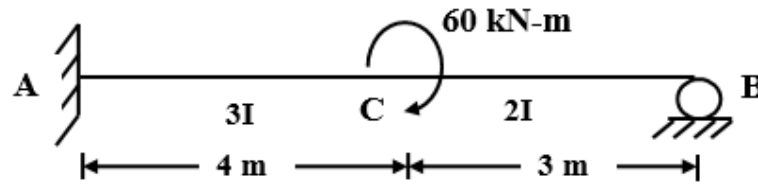


Figure: 8

OR

- (b) Analyse the beam shown in **Figure 9** by consistent deformation method. Draw shear force and bending moment diagrams. [07]

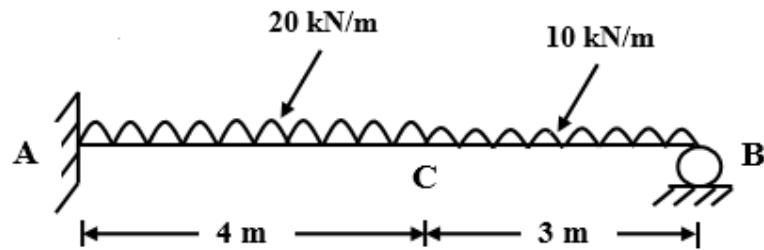


Figure: 9
